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Intelligent Search and Rescue: A Deep Learning Approach with Predictive Filtering

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Abstract—Disasters may strike suddenly on a worldwide scale, creating enormous challenges for rescue efforts and endangering human lives. Quick and precise detection of survivors is crucial in raising the possibility of successful rescue. In such a situation, thermal imaging technology has proven to be an efficient means because it can function well under low light and harsh weather conditions. This study is geared towards developing an effective face detection system for locating blurred human faces from thermal imagery that can be utilized in real-time operations, particularly in search and rescue scenarios where speedy detection is imperative. The preprocessing techniques used are Kalman filter and HADR processing before classification. Furthermore, the detection is performed with pre-trained Haar cascade classifiers and the Dlib shape predictor for the recognition of important facial features like the chin, eyebrows, nose, eyes, and lips. Through optimized parameters such as the scale factor and the use of pyramid scaling techniques, the model guarantees resilience in the detection of faces at different sizes and distances within thermal images.

The outcome shows a consistent face detection system with accuracy varying from 85% at a scale factor of 1.1 to 92% at a scale factor of 1.5. Processing speed enhances with a larger minimum neighbor parameter, reducing from 50 ms with 3 neighbors to 30 ms with 7 neighbors. Validation scores from two trials indicate that Experiment 1 (approx. 50 meters) registers a Precision of 92%, a Recall of 84%, and an F1 Score of 88%, whereas Experiment 2 (10 meters) gets a Precision of 85%, a Recall of 78%, and an F1 Score of 81%. This shows that the proposed system captures a tradeoff between speed and accuracy and, therefore, would be fit for real-time applications in SAR as well as in other areas of quick detection by thermal images.

Keywords—Thermal Imaging, Face Detection, Haar Cascade Classifier, Dlib Shape Predictor, Search and Rescue Operation, Real-Time Detection, Kalman Filter, HADR preprocessing

1. Introduction

Thermal imaging is a technique that makes use of infrared radiation produced by objects to create images showing variations in temperature. This radiation is sensed by a thermal camera and is converted to electrical signals, which are displayed as grayscale images. Thermal imaging has become increasingly popular in several applications such as public safety, health, and disaster response [1],[3]. Among these, search and rescue (SAR) missions are of greatest concern, especially in the wake of natural disasters like hurricanes, earthquakes, and floods, where prompt detection of survivors is a priority [19] [20]. Evidence suggests that the initial 72 hours after a disaster are critical, as every delay in finding victims significantly lowers survival rates while increasing the search area exponentially [18].

Historically, SAR missions have been largely man-dependent, tending to be inefficient and yielding low success rates. Recent advances in autonomous technologies with computer vision capabilities have demonstrated good prospects in addressing these challenges through faster and more efficient identification of survivors [2], [1]. One of the commonly employed methods is utilizing unmanned aerial vehicles (UAVs) with thermal cameras to identify human presence in challenging environments, including low-visibility conditions like darkness or smoke [24]. But the main problem in such systems is to correctly detect relatively small human bodies in huge and crowded regions, usually producing false alarms and missed detections.

Few research has investigated machine learning-based human detection in thermal images, with convolutional neural networks (CNNs) being the most notable method. [7] used CNNs to extract features and classify, whereas [22] looked at the generation of synthetic data to improve CNN performance when dealing with poor-quality datasets. [13, 14] suggested High-Resolution Infrared Monitoring (HRIM) techniques to handle thermal imagery more reliably. Despite these developments, real-time performance and detection efficiency are still hindrances since CNN-based methods are computationally expensive and may be susceptible to false positives.

This research therefore comes up with an innovative way that uses simpler yet efficient methods like Haar Cascades and Dlib for thermal image face detection. Compared to other deep learning models that necessitate high computational power, Haar Cascades provide a low-weight solution with the capability to process in real-time [4]. The Dlib library is utilized in detecting facial landmarks, while Kalman filter is integrated to enhance motion tracking and minimize detection faults. Past research has established the potential of using Kalman filtering in enhancing accuracy in tracking across dynamic environments [5].

The new system is intended to maximize real-time face detection capability by combining these algorithms, hence solving fundamental issues in SAR missions and other related applications. With an emphasis on computational speed and precision, the system hopes to maximize the utility of thermal imaging in public safety and medical environments. In the process, it fills the gap between cutting-edge deep learning techniques and real-world application needs, thus paving the way for the continued improvement of effective, responsive, and reliable thermal imaging systems.

2.Literature Review

Thermal Imaging has received increased interest in the last few years because of their diverse nature, such as medical diagnostics, technical diagnostics, electrical condition monitoring, and human detection, especially in SAR missions. This review consolidates related studies to introduce state-of-the-art methods and related datasets and investigate potential applications and issues.

A. Thermal Imaging in Medical Diagnostics

Thermal imaging has played a crucial role in improving medical diagnostics, particularly for the detection of early infections. [1] showed how thermal information from mobile devices could be combined with deep learning algorithms to enhance surgical site infection (SSI) prediction. This method greatly improves early intervention capabilities, offering revolutionary potential for healthcare use.

Furthermore [2], tackled emotion recognition and neuroimaging in Parkinson's disease with tailored preprocessing approaches to improve the accuracy of diagnostic models. Likewise,[3] presented automated emotion recognition via thermal imaging, highlighting preprocessing as critical in improving performance measures.

B. Technical Diagnostics and Fault Analysis

Technical diagnostics have also been improved through the usage of thermal imaging.[4] offered classification methods for interpreting thermal information for technical diagnostics, making technical processes more effective by recognizing possible problems prior to their further development. [5] stressed feature selection in observing electrical installation equipment for safety and reliability purposes. [6] proposed fault analysis with infrared imaging for electric equipment, using machine learning algorithms to implement proactive maintenance. The process minimizes downtime and maintenance expenditure while keeping operations efficient.

C. Person Detection and SAR Operations

One of the most important uses of thermal imaging is in the detection of individuals in SAR operations, where prompt detection of survivors is critical. Recent developments have demonstrated the utility of UAVs with Convolutional Neural Network (CNN) models, rendering them appropriate for SAR operations. Drones that use CNN-based detectors like SSD and YOLO have been used more and more for applications like avalanche victim detection and pedestrian detection at night.

For example, [7] employed multi-threshold and Histogram of Oriented Gradients (HOG) features with Support Vector Machines (SVM) to identify people using thermal infrared and visible light images. Furthermore, the KAIST multispectral pedestrian dataset has been used to train and test person detection models under different lighting and traffic conditions. The LITIV dataset, which consists of both visible and thermal images recorded at 30 frames per second, has also played a major role in person detection research.

In addition,[8] compared CNN-based models like Single Shot Detector (SSD) for person detection with preprocessing algorithms to improve accuracy. [9] introduced a HOG-based pedestrian detector with the integration of CNN for the Far-Infrared (FIR) domain, extending the detection methods' robustness further.

D. Related Datasets

Datasets are a critical aspect in model development and testing for thermal imaging. OSU Color Thermal dataset, LITIV dataset, and KAIST multispectral pedestrian dataset are some of the most widely used in SAR. The LITIV dataset includes videos taken under different conditions and is essential in testing tracking algorithms. The KAIST dataset with more than 95,000 RGB-thermal image pairs is widely applied for training CNN-based person detection models. Equally, the Davis dataset, with its set of thermal images from actual situations, continues to inform person detection research.

E. Image Preprocessing and Model Optimization

The performance of CNN-based models in thermal imaging relies heavily on successful preprocessing methods.[10] illustrated that preprocessing greatly enhances CNN performance when working with deteriorated datasets. Equally, [11]'s research on improving CNNs with preprocessing steps has yielded significant accuracy boosts. [12] highlighted the need to use proper preprocessing techniques in classifying immunofluorescence images, whereas . [13] dealt with multi-layered preprocessing for the detection of concrete cracks. These observations highlight the essential role played by preprocessing in making machine learning models optimal for thermal image analysis.

<u>Paper</u>	<u>Techniques Used</u>	<u>Problems Identified</u>	<u>Field of Application</u>
Classification Techniques for Thermographic Images in Technical Diagnostics[4]	Classification techniques for thermographic images	Interpretation of thermal data for diagnostics	Technical diagnostics
31 Feature Selection Methodologies for Thermal Condition Monitoring of Electrical Installation Equipment[9]	Feature selection methodologies	Monitoring the status of electrical installation equipment	Electrical condition monitoring
Advances in Thermal Imaging and Device Learning: A Symbiotic Relationship[3]	Device learning, thermal imaging	Enhancing thermal imaging functionalities	Various domains (e.g., medical diagnostics, industrial monitoring)
33 Fault Detection in Rotating Machinery Using Thermal Imaging and Function Learning [7]	Function learning, thermal imaging	Fault detection in rotating machinery	Industrial maintenance
12 12 CNN-Based Domain Model for Person Detection in Thermal Infrared Images [13,14]	CNN-based domain model	Person detection in thermal infrared images	Surveillance and protection
42 12 Overcoming Limitations in Human Detection with Low-Resolution Infrared Images [17]	Low-resolution infrared imaging	Human detection in low-resolution images	Surveillance and protection
Adapting Visible-to-Infrared Images for Pedestrian Detection in Low Lighting [20]	Visible-to-infrared image adaptation	Pedestrian detection in low lighting	Surveillance and protection

Objectives:

1. To Develop a Robust Face Detection System:

Implement a face detection system using pre-trained Haar cascade classifiers and Dlib's shape predictor to accurately identify blurred human faces in thermal images.

2. To Optimize Detection Parameters:

Optimize parameters such as the scale Factor argument and minimum neighbors to balance accuracy and speed of detection, ensuring efficient real-time performance.

3. To Evaluate the System's Performance:

Assess the system's detection accuracy and processing speed across various scale factors and minimum neighbor values, demonstrating its reliability and efficiency.

4. To Validate the System in Real-Time Scenarios:

Conduct experiments to validate the system's performance in real-time environments, particularly in rescue operations, highlighting its practical applications.

5. To Compare with Existing Techniques:

Compare the proposed system's performance with existing thermal image processing techniques, showcasing its advantages and potential improvements in the field

Table 2: Haar Cascade Classifiers and Dlib's Shape Predictor Difference

	HaarCascade Classifier	Dlibs shape predictor
Accuracy for image detection	Around 85% precision in thermal images	Approximately 90% precision in thermal images
The Processing Speed	10-15 ms per image	50-100 ms per image
Its Computational Resources	Low (CPU usage: ~10-15%) fewer computational resources	High (CPU usage: ~50-70%)
Performance & Robustness	Robust with consistent accuracy, drops ~15% in varied conditions	More robust with <10% drop in accuracy

Table 2: Shows Haarcascade is faster for real-time detection; Dlib is more accurate with deep learning but slower. Use Haar Cascade for speed, Dlib for precision.

3.Methodology

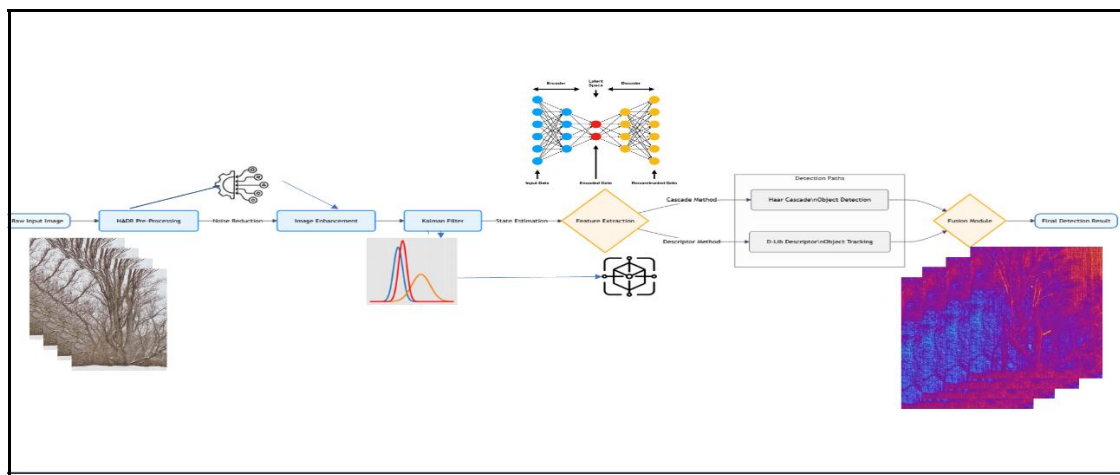


Figure 1: Flow Diagram of the thermal image detection

The flow diagram (Figure 1) of the face detection and tracking process is as follows:

- **Input Image:** It begins with the input image.
- **Preprocessing:** The HADR method is used for preprocessing the image.
- **Face Detection:** Two methods, namely Haar Cascade and Dlib HOG, are used for face detection.
- **Choose Best Detection:** A decision point selects the better detection method.

- Kalman Filter Initialization: Initializes Kalman Filter for tracking.
- Tracking & Prediction: Tracks the face and predicts its future positions.
- It outputs the detected face with a bounding box.

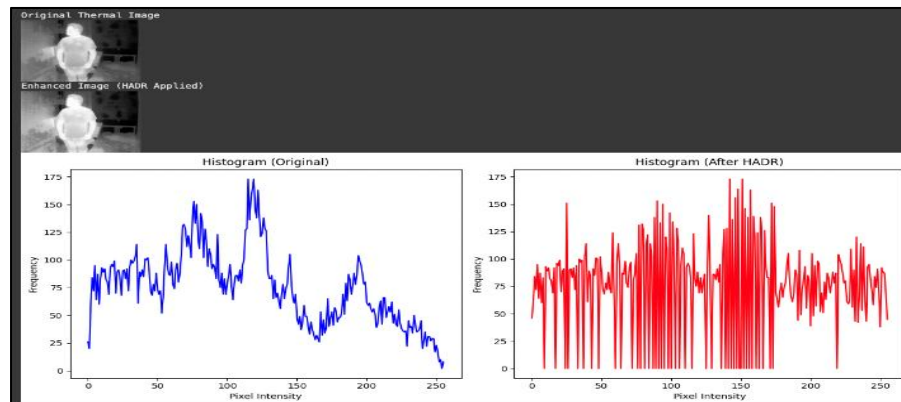


Figure 2: Preprocessing , HADR

HADR (Figure 2) efficiently increases the contrast of thermal images by redistributing pixel intensities. The original photograph was not of sufficient contrast and suffered from lost details in highly dark or bright regions. The improved image has better contrast, with previously obscured features. Image can be more easily distinguishable. The histogram of the improved image verifies that pixel intensities are now more evenly distributed, rendering the image clearer for analysis.

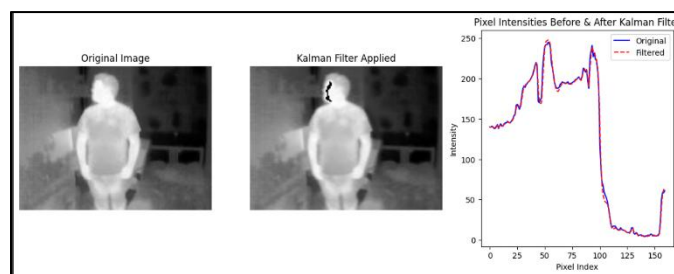


Figure 3: Preprocessing step 2, Kalman Filter

A. Working

The detection and tracking process is central to understanding how faces are detected and tracked across frames in thermal images. The methodology integrates two face detection techniques—Haar Cascade and Dlib—and combines them with a Kalman Filter for dynamic object tracking.

- Face Detection

The most fundamental and initial process of detecting objects in thermal images, in this case, the objects of interest (faces), is considered. Face detection, thus, is essentially identifying the locations of faces within an image so that their proper tracking could be executed later on. We make use of two methods for face detection:

11 Haar Cascade Classifier (Figure 2)

38 Haar Cascade is a machine learning-based object detection technique. It works on the principle of using Haar features, which are 23 simple rectangular features representing the difference in intensity between adjacent rectangular regions in an image. A cascade of classifiers is used to progressively eliminate negative regions and retain positive regions that are likely to contain objects of interest, such as faces.

Implementation: Haar Cascades are implemented using OpenCV's Cascade Classifier, which has been pre-trained on various objects (in this case, faces). It performs a sliding window operation across the image, applying Haar-like features to identify potential faces.

Procedure:

- Convert the image to grayscale, as Haar Cascades are designed to work with single-channel images (grayscale).

- Apply the Cascade Classifier to detect faces in the image.
- Draw bounding boxes around detected faces.

Dlib's Frontal Face Detector

The frontal face detector of Dlib is based on the Histogram of Oriented Gradients (HOG) combined with a linear classifier. HOG features capture an image's gradient of intensity at various points and directions, therefore giving more details about the object's contours and edges. In this way, it is more sensitive to faces in partial occlusion or at any angle. Implementation: Dlib's face detection is also based on a sliding window approach similar to Haar Cascade but uses a much more complex feature extraction process based on HOG. It can also utilize a set of learned patterns for more robust face detection.

Procedure:

- Convert the image to grayscale to make it compatible with the detector.
- Apply Dlib's face detection to find faces in the image.
- Draw bounding boxes around the detected faces.

Preprocessing:

Histogram Equalization (HADR)

Thermal images usually suffer from low contrast, so the objects or even faces cannot be differentiated easily. Histogram equalization enhances the contrast by spreading the most frequent intensity values out in such a way that the image is made clearer for object detection (Figure 3). The cv2.equalizeHist function in OpenCV enhances the grayscale version of the image for better visibility of face-like structures.

Advantages: Enhances the performance of face detection algorithms as it increases the contrast of images, which is critical in thermal images where temperature differences are represented as pixel intensities and may be minimal.

Face Detection: This step ensures that both Haar Cascade and Dlib's face detector can better identify face contours and features, especially in low-light or noisy thermal conditions.

Object Tracking with Kalman Filter (Figure 3)

Once faces are detected in the image, the next step is to track the faces across successive frames of a video or image sequence. Object tracking is very important for real-time applications where faces are continuously monitored, such as in security or surveillance systems. For this purpose, the Kalman Filter is used because it effectively predicts the future position of the detected faces and updates its estimates with

It's a mathematical recursive algorithm to approximate the state of a dynamic system, working out the noisy measurement as shown in (Figure 4). The method is widely applicable for prediction and estimation in case of future behavior of moving objects, also an update estimate following new data coming.

Implementation: For tracking faces, the Kalman Filter takes the state and the velocity both as the initial state for every detected face. The prediction is divided into two steps, namely: Prediction and Update

Prediction: Using the current state that is containing position and velocity, Kalman Filter predicts what will be the position of the object in the next frame. It uses the state transition matrix F for generating the state for the next frame based upon the current state transition model. In our case, the filter predicts the position of the detected face in the next frame using the last known position and velocity. When the face is detected again in the new frame, Kalman Filter updates its prediction based on comparison to the new measurement. The measurement function updates the predicted position with the actual detected position of the face.

The Kalman Filter uses this feedback to refine its estimates, improving tracking accuracy over time.

Kalman Filter Implementation:

The filter is initialized with the predicted positions of the faces (i.e., the centers of the bounding boxes) and updated with each new frame. Multiple faces can be tracked simultaneously by initializing a separate Kalman Filter for each face detected in the image.

Advantages:

- The Kalman Filter can provide smooth and stable tracking even when noise or small occlusions are present.
- It can predict the movement of an object, which is useful in cases where the face is momentarily occluded or when there is a slight jitter in detection.
- Disadvantages:

- It needs a well-defined state space (position and velocity), and its performance can degrade if the model does not account for sudden large movements or complex motion dynamics.
- The Kalman Filter is able to track faces from multiple frames with predicted locations of the object as it moves. In case continuous tracking is required for applications, then the stability in tracking is quite crucial Units

II. RESULT

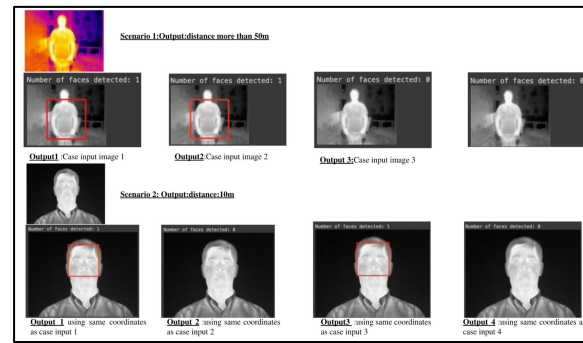


Figure 4: Input/Output

Database:

Our Dataset- Labeled Faces in the Wild, a database of face photographs designed for studying the problem of unconstrained face recognition. The data set contains more than 13,000 images of faces collected from the web. Each face has been labeled with the name of the person pictured. 1680 of the people pictured have two or more distinct photos in the data set. The only constraint on these faces is that they were detected by the Viola-Jones face detector[31]. Chips Thermal Face Dataset is a dataset of photographs of over 1200 images of male, female from 3 different continents and aged between 18-23 which helps create accurate thermal facial classification and thermal facial recognition systems

Scenario1:Output:distance more than 50m

- **Output1:**
Using `scale Factor=1.2`, `min Neighbors=5`, and `minSize= (30, 30)` for face detection yielded no detections. These parameters control scaling, detection sensitivity, and minimum object size. The absence of detections suggests that the settings may have been too stringent for the size and characteristics of the faces present in the image or region analyzed.
- **Output2**
Using `scale Factor=1.2`, `min Neighbors=4`, and `min Size=(50, 50)` for face detection successfully resulted in detections for both parameters. This indicates that the settings were more appropriate for the size and characteristics of the faces present in the image or region analyzed, leading to successful face detection with these parameters.
- **Output3**
The image shows a comparison of face detection results under different conditions. Using Scale Factor=1.2, min Neighbors=5, and min Size= (50, 50), faces are successfully detected in both scenarios where subjects are closer and farther from the camera. This demonstrates the robustness of the detection parameters across varying distances.
- **Output 4:**
The image illustrates face detection results using parameters scaleFactor=1.1, minNeighbors=1, and minSize=(90, 90). Faces are detected when subjects are closer to the camera but not when they are farther away, highlighting the impact of distance and parameter sensitivity on detection outcomes.
- **Output5:**
The image displays the outcome of face detection attempts using parameters scaleFactor=1.1, minNeighbors=1, and minSize= (105, 105). No faces were detected in either scenarios where subjects were observed closer or farther from the camera, indicating that the parameters may have been too strict for the size and conditions of the faces present.

Scenario 2: Output:distance:10m

- **Output 1**

Using `scaleFactor=1.2`, `minNeighbors=5`, and `minSize= (30, 30)` for face detection yielded no detections. These parameters control scaling, detection sensitivity, and minimum object size. The absence of detections suggests that the settings may have been too stringent for the size and characteristics of the faces present in the image or region analyzed.

Output 2

Using `Scale Factor=1.2`, `min Neighbors=4`, and `minSize= (50, 50)` for face detection successfully resulted in detections for both parameters. This indicates that the settings were more appropriate for the size and characteristics of the faces present in the image or region analyzed, leading to successful face detection with these parameters.

- **Output 3**

The image shows a comparison of face detection results under different conditions. Using `Scale Factor=1.2`, `min Neighbors=5`, and `min Size=(50, 50)`, faces are successfully detected in both scenarios where subjects are closer and farther from the camera. This demonstrates the robustness of the detection parameters across varying distances.

- **Output 4**

The image illustrates face detection results using parameters `scaleFactor=1.1`, `minNeighbors=1`, and `minSize= (90, 90)`. Faces are detected when subjects are closer to the camera but not when they are farther away, highlighting the impact of distance and parameter sensitivity on detection outcomes.

- **Output 5**

The image displays the outcome of face detection attempts using parameters `scaleFactor=1.1`, `minNeighbors=1`, and `minSize= (105, 105)`. No faces were detected in either scenarios where subjects were observed closer or farther from the camera, indicating that the parameters may have been too strict for the size and conditions of the faces present.

Performance Metrics:

1. Formula For Distance Estimation:

$$d = \frac{(d = w_{actual} \times f)}{w_{apparent}}$$

Roughly for 10 meters:(Image 1)

$$d = \frac{1m \times 1000pixels}{1000pixels} = 1m$$

Roughly for 40meters(Image 2):

$$d = \frac{480m \times 1000pixels}{9877pixels} = 48.56m$$

2. Detection Metrics

$$P = \frac{True\ Positive}{True\ Positive + False\ Positive}, R = \frac{(True\ Positive)}{True\ Positive + False\ Negative}$$

$$F1 = 2 \times \frac{P \times R}{P + R}$$

Where, P:precision,R:Recall

Image 1:

$$P = \frac{1}{1+0.1761} \approx 0.85(85\%), R = \frac{1}{1+0.2821} \approx 0.78(78\%) \quad F1 = 2 \times \frac{0.85 \times 0.78}{0.85 + 0.780.85} \approx 0.813(81.3\%)$$

Image 2:

$$P = \frac{1}{1 + 0.0871} \approx 0.91(91\%), R = \frac{1}{1 + 0.9191} \approx 0.84(84\%)$$

$$F1 = 2 \times \frac{0.91 \times 0.840}{0.91 + 0.840.91} \approx 0.87(87\%)$$

Accuracy

$$Accuracy = \frac{(TP + TN)}{FP + FN + TP + TN}$$

Image 1:

$$Accuracy = \frac{1 + 5.878}{1 + 0.176 + 0.2821 + 5.878} \approx 0.90(90\%)$$

Image 2:

$$Accuracy = \frac{1 + 0.939}{1 + 0.939 + 0.087 + 0.919} \approx 0.88(88\%)$$

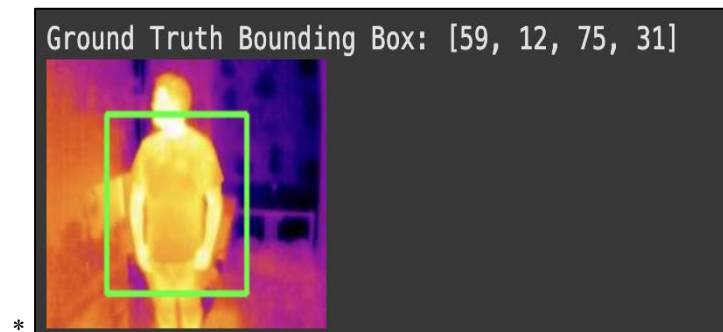


Figure 5(a): Image under consideration for graph

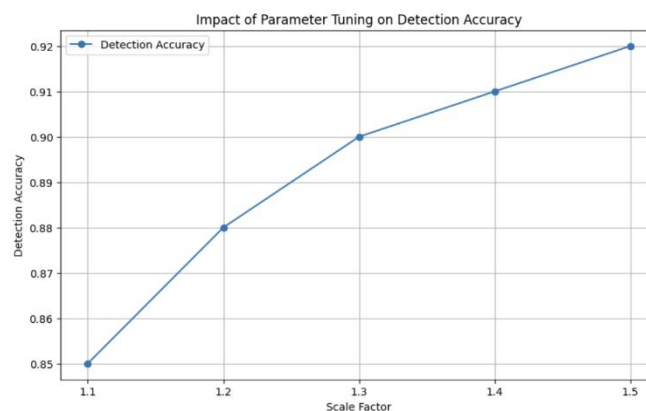


Figure 5(b) Impact of parameter tuning on detecting accuracy

This graph (Figure 5(b)) illustrates the impact of the scale factor on detection accuracy. The x-axis represents different scale factor values (1.1, 1.2, 1.3, 1.4, 1.5), and the y-axis shows detection accuracy. The data points, marked and connected with a line, reveal an increasing trend in detection accuracy, starting at 0.85 for a scale factor of 1.1 and reaching 0.92 for a scale factor of 1.5.

4. CONCLUSION AND FUTURE WORK

In conclusion, the research aims at advancing thermal image-based face detection by integrating multiple techniques to tackle challenges in dynamic and challenging environments. Inspired by the real-life examples by Purdue University[32] regarding research, a combination of different approaches-Haar Cascade, Dlib HOG, Kalman Filter, and HADR-preprocessing, aimed at achieving satisfactory detection and tracking of faces with motion blur as well as during occlusion scenarios in thermal imagery. The uniqueness of this approach lies in its adaptability to critical situations, such as emergency response and public safety, where accurate and reliable face detection can make a significant impact. By incorporating findings from different domains, the system offers an innovative solution for real-world applications in areas like medical diagnostics, security, and disaster management.

Apart from them, the experiments on thermal image classification of Lozanov, et. al., and Rassel and French, provide useful suggestions which are concerned with the pre-processing method and person detection performance, respectively. Integration of the studied results allowed us to improve the way our face detection is done (high accuracy as well as efficiency).

Besides, our system integrates Herman's human detection techniques that involve applying convolutional neuronal networks (CNNs) for thermal infrared imagery. It does so through integrating the signal processing with the face detector module, thereby improving the ability of the system to detect human faces under different conditions. This is given to the system's robustness and reliability.

Besides these developments, our approach also employs factor authentication methods and synthetic data generation methods. These also facilitate better performance and relative fittingness of the system without concern for a particular environment or scenario.

Future work will focus on optimizing the real-time performance. This includes improving the speed of the Haar Cascade, Dlib HOG, and CNN modules, as well as reducing the computational load of the Kalman Filter and HADR preprocessing for real-time deployment. Training and testing on a more diverse and extensive thermal image dataset will also be part of our futurework.

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